



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

Course Code: SE 200

Class: BSCS-13E

Software Engineering

Instructor: Sir Sarosh Tahir

Project Deliverable 2

Team Members

Name	CMS
AHMED RAZA	465607
BILAL RANA	454035
MEERAB CHAUDRY	477284
M. SAMAMA USMAN	454520



1. Functional Requirements:

The following functional requirements define what the Moodify system must do. Each requirement is clear and testable.

FR-01	User Registration & Authentication	The system shall allow users to register with email/password or third-party OAuth (Google). Users shall be able to log in, log out, and reset passwords securely.
FR-02	Mood Logging	Users shall be able to log their current mood by selecting from a set of icons/emojis representing mood levels (1–5 scale). The logging flow shall complete in under 30 seconds.
FR-03	Emotion & Activity Tagging	Users shall be able to attach emotion tags (e.g., anxious, happy, tired) and activity tags (e.g., exercise, sleep, social) to each mood entry.
FR-04	Optional Journal Note	Users shall be able to add a free-text note of up to 500 characters to each mood log entry as an optional field.
FR-05	Mood History & Calendar View	Users shall be able to view their past mood entries in a calendar/timeline format showing mood color per day (Year in Pixels style).
FR-06	Analytics Dashboard	The system shall generate mood trend charts (weekly, monthly, yearly) and correlate mood scores with tracked activities to surface patterns and insights.
FR-07	AI-Powered Personalized Suggestions	Based on mood patterns and activity data, the system shall generate context-aware wellness suggestions such as breathing exercises, journaling prompts, or activity recommendations.
FR-08	CBT Tools & Thought Records	The system shall provide a Cognitive Behavioral Therapy (CBT) thought record feature, guiding users through identifying negative thoughts and reframing them with structured prompts.



FR-09	Breathing Exercises	The system shall offer guided breathing exercises (e.g., box breathing, 4-7-8 breathing) accessible for free from the home screen without requiring a premium subscription.
FR-10	Gratitude Journal	Users shall be able to write daily gratitude entries (3 items minimum prompt) stored separately from mood logs with their own history view.
FR-11	Clinical Assessments (PHQ-9 & GAD-7)	The system shall provide validated clinical questionnaires (PHQ-9 for depression, GAD-7 for anxiety) with scored results, trend tracking, and a disclaimer to consult a professional.
FR-12	Gamification – Streaks & Achievements	The system shall track and display daily logging streaks, award achievement badges for milestones (e.g., 7-day streak, 30 logs), and show a progress summary on the dashboard.
FR-13	Therapist Report Generation	Users shall be able to generate a shareable mood and assessment summary report (PDF format) to share with their therapist or counselor.
FR-14	Data Export	Users shall be able to export all mood entries and assessment results in CSV and PDF formats at any time, free of charge, including an option to permanently delete all their data.
FR-15	Push Notification Reminders	The system shall allow users to configure customizable daily reminder notifications to log their mood at a preferred time, with the option to enable or disable them.



2. Non-Functional Requirements:

Non-functional requirements define the quality attributes and operational constraints that Moodify must satisfy. These are critical for ensuring the system is safe, trustworthy, and scalable, particularly given that it handles sensitive personal health data.

NFR-01	Performance	The app shall load the mood logging screen within 2 seconds on a standard 4G connection. The analytics dashboard shall render within 3 seconds for up to 365 data points.
NFR-02	Security & Privacy	All user data shall be encrypted at rest using AES-256 and in transit using TLS 1.3. The app shall implement end-to-end encryption for mood entries and personal health data.
NFR-03	Usability	The app shall achieve a System Usability Scale (SUS) score of at least 75 (Good) in user testing. The mood logging flow shall require no more than 3 taps to complete a basic entry.
NFR-04	Availability & Reliability	The backend services shall maintain 99.5% uptime per month. The app shall support offline-first operation, allowing mood logging without an internet connection and syncing when connectivity is restored.
NFR-05	Scalability	The system architecture shall support up to 100,000 concurrent users without degradation in performance, using cloud-based autoscaling for the backend infrastructure.
NFR-06	Compatibility	The app shall be compatible with Android 8.0 (API Level 26) and above, and iOS 14.0 and above. UI shall be responsive across screen sizes from 5" to 7".
NFR-07	Data Retention & Compliance	User data shall be retained only as long as the account is active. Upon account deletion, all personal data shall be permanently purged within 30 days in compliance with GDPR principles.
NFR-08	Maintainability	The codebase shall maintain at least 70% unit test coverage. All modules shall follow a clean architecture pattern to allow feature additions without breaking existing functionality.



3. Assumptions

The following assumptions have been made during requirements elicitation. These are conditions believed to be true for the purposes of this project but have not been formally verified. If any assumption proves incorrect, the corresponding requirements may need to be revised.

3.1 User & Environment Assumptions

- Users have access to a modern web browser (Chrome, Firefox, Safari, or Edge) on any device ,desktop, laptop, or smartphone , capable of running a Progressive Web App.
- Users have sufficient device storage (at least 50 MB) to support local PWA caching and offline data storage via the browser.
- The primary target users (ages 16–35) are sufficiently comfortable with web applications to navigate mood logging and analytics features without in-person training.
- Users are assumed to interact with the app at least once per day during active use; the notification and streak system is designed around this frequency.
- Users have access to an internet connection at least once per day for background cloud sync, even though core features function offline.

3.2 Clinical & Content Assumptions

- Moodify is not a medical device and does not provide clinical diagnosis. It is assumed users understand this distinction through onboarding disclaimers.
- PHQ-9 and GAD-7 assessments are used for self-monitoring only. It is assumed users will seek a licensed professional if their scores indicate moderate or severe levels.
- AI-generated wellness suggestions are assumed to be supplementary and non-prescriptive; users are not expected to follow them without personal judgment.

3.3 Technical Assumptions

- The backend will be hosted on a cloud provider (e.g., AWS or Firebase) that supports autoscaling. It is assumed cloud costs remain within budget for up to 50,000 users in the first six months.
- Push notifications will be delivered via the Web Push API, which is assumed to be supported across all target browsers. It is assumed users will grant notification permissions during onboarding.
- Third-party OAuth providers (Google) will maintain API compatibility throughout the development lifecycle without breaking changes.
- Users who choose to install the PWA on their device home screen are assumed to be on a browser version that supports PWA installation (Chrome 73+, Safari 16.4+, Firefox 113+).



4. Constraints

Constraints are fixed limitations that the project must operate within, regardless of design decisions. Unlike assumptions, constraints are known and non-negotiable boundaries imposed by technical, regulatory, time, resource, or business factors.

4.1 Technical Constraints

- The app must be built as a **web application** accessible via modern browsers (Chrome, Firefox, Safari, Edge). Native mobile apps for Android and iOS are out of scope for version 1.
- The application must not depend on any proprietary AI model that cannot be accessed via a publicly available API, to ensure maintainability and cost predictability.

4.2 Regulatory & Privacy Constraints

- All personal health data must be handled in compliance with GDPR principles, including the right to access, portability, and erasure. Full data deletion must be available in-app.
- Clinical assessment content (PHQ-9, GAD-7) must be used in accordance with the respective licensing terms and must always include a disclaimer that results are not a clinical diagnosis.
- The app must not collect or share any user data with third-party advertisers. All analytics used internally for product improvement must be anonymized and aggregated.

4.3 Time & Resource Constraints

- The Minimum Viable Product (MVP) must be delivered within the academic semester timeline. Features marked as "Planned (v2)", including Medication Tracking and the Nervous System Regulation Tool, are explicitly deferred to a post-MVP release.
- The development team consists of four members with existing academic commitments; the total available development hours per week are limited, and scope must be managed accordingly.

4.4 Business Constraints

- Core features, including mood logging, basic analytics, breathing exercises, and data export, must remain free of charge to all users. Monetization must only apply to advanced or premium features.
- The app's premium pricing tier must remain competitive with existing apps in the space (Daylio at Rs.1400/month, Moodfit at Rs.2790/month) and may not exceed Rs.800/month at launch without explicit review.



Use Case Descriptions

This document covers five use cases for Moodify. Each one describes a core feature of the app with the actors involved, preconditions, the main step-by-step flow, alternative paths, and what happens when things go wrong.

UC-01: Log a Mood Entry

Use Case ID	UC-01
Name	Log a Mood Entry
Actors	End User (primary), Moodify System (secondary)
Description	The user logs their current mood by picking a mood level, tagging emotions and activities, and optionally writing a note. The whole thing should take under 30 seconds.
Trigger	User taps the "+" button on the home screen, or taps a daily reminder notification.
Preconditions	App is installed and onboarding is done. Device has enough local storage.
Postconditions	Mood entry saved locally with a timestamp. Streak counter updated. Entry shows up in history and analytics.

Main Flow:

1. User opens the app or taps the reminder notification.
2. Home screen shows today's date, streak count, and a "+" button.
3. User taps "+" and the mood selection screen appears with 5 emoji levels (Awful to Great).
4. User picks a mood level.
5. System shows a grid of emotion tags (Anxious, Happy, Tired, Calm, etc.). User taps the ones that apply.
6. System moves to the activity screen. User picks what they did today (Exercise, Work, Social, Sleep, etc.).
7. An optional note field appears. User can type something or attach a photo, or just skip it.



National University of Sciences and Technology (NUST) School of Electrical Engineering and Computer Science

8. User taps Save.
9. System validates (mood level is required), saves the entry, updates the streak, and shows a quick confirmation animation.
10. Home screen now reflects today's logged mood.

Alternative Flows:

A1: User skips emotion tags or activities. Both are optional. The system just moves to the next screen and saves whatever the user did select. The minimum valid entry is just a mood level.

A2: User logs multiple entries in one day. The app allows it and saves each one with its own timestamp. Analytics will average them for the daily score.

A3: User edits a previous entry. They tap it from the history/calendar, the app opens it in edit mode with everything pre-filled, and they can change whatever they want and hit Update.

Exception Flows:

E1: Storage is full. System shows an error but keeps the entry in memory so the user does not lose it.

E2: App crashes mid-entry. The system auto-saves a draft. When the user reopens the app, it asks if they want to continue or discard the draft.

E3: User tries to save without selecting a mood. System highlights the mood selection and shows a message asking them to pick one.



UC-02: View Mood Insights and Analytics

Use Case ID	UC-02
Name	View Mood Insights and Analytics
Actors	End User (primary), Analytics Engine (secondary)
Description	User checks their mood trends, activity correlations, and AI-detected patterns over different time periods (week, month, year).
Trigger	User taps the Insights tab, or taps a weekly summary notification.
Preconditions	At least 3 mood entries exist (minimum for generating trends).
Postconditions	User has viewed their analytics. Detected patterns are flagged for the suggestion engine.

Main Flow:

1. User taps the Insights tab.
2. Dashboard loads with the weekly view showing a mood trend line, average mood score, mood distribution (e.g., 40% Good, 30% Great), and streak count.
3. User scrolls to the activity correlation section. A bar chart ranks activities by their average associated mood (e.g., Exercise = 4.2 avg, Overtime = 2.1 avg).
4. User taps a specific activity to see a filtered view with only entries where that activity was logged.
5. User switches the time period to Month. The dashboard reloads with a calendar heatmap and updated charts.
6. At the bottom, a Patterns Detected section shows AI observations like "Your mood drops on Mondays" or "You feel better on days you exercise."
7. User taps a pattern to see the supporting data.

Alternative Flows:

A1: Fewer than 3 entries. Instead of empty charts, the system shows a friendly message like "Keep logging! We need a few more entries to show your trends" with a progress indicator.

A2: User opens Year in Pixels view. A grid of 365 colored dots appears, one per day, colored by mood. Gray dots mean no entry. Tapping a dot shows that day's details.



A3: User checks emotion frequency. An Emotions tab shows a ranked list of most tagged emotions and how each one correlates with overall mood.

Exception Flows:

E1: No data for selected period. System shows a message saying no entries were found and suggests picking a different time range.

E2: Analytics engine error. System shows a retry button and suggests contacting support if the problem keeps happening.



UC-03: Receive Personalized Wellness Suggestions

Use Case ID	UC-03
Name	Receive Personalized Wellness Suggestions
Actors	End User (primary), AI Recommendation Engine (secondary)
Description	After logging a mood, the system looks at recent patterns and offers context-aware suggestions like breathing exercises, journaling prompts, or activity changes.
Trigger	Automatically triggered after saving a mood entry, or when user opens the Suggestions section manually.
Preconditions	At least 5 entries exist for the AI to generate meaningful suggestions. User has not turned off suggestions in settings.
Postconditions	User has seen 1 to 3 personalized suggestions. If the user engaged with one, the interaction is logged for improving future recommendations.

Main Flow:

1. User finishes logging a mood entry.
2. System analyzes the entry in context of the last 7 days: mood level, trend direction, tagged emotions, activities, time of day.
3. System generates 1 to 3 suggestions. For example, if mood is Bad and Anxious is tagged, it might suggest a breathing exercise, a journaling prompt, and a reminder that physical activity helped in the past.
4. Suggestions appear as cards with a title, short description, estimated time, and a Try It button.
5. User taps Try It on the breathing exercise.
6. Built-in breathing module opens with an animated guide (inhale, hold, exhale) and a timer.
7. User finishes. System asks "Did this help?" with thumbs up/down.
8. User taps thumbs up. System logs the feedback and prioritizes similar suggestions in the future.

Alternative Flows:

A1: User dismisses all suggestions. System respects it and does not show them again until the next entry or until the user visits Suggestions manually. Dismissals are logged so the system learns.



National University of Sciences and Technology (NUST) School of Electrical Engineering and Computer Science

A2: User picks the journaling prompt instead. System opens a text editor with the prompt as a header. The response gets linked to that day's mood entry.

A3: Fewer than 5 entries. AI cannot personalize yet, so it shows generic wellness tips with a note saying suggestions will get more personal as the user logs more.

Exception Flows:

E1: Suggestion module fails to load. System shows an error and offers to try a different suggestion from the remaining cards.

E2: User gives thumbs down on 5+ suggestions in a row. System asks what kind of support they actually find helpful and shows a short preference survey.



UC-04: Complete a Clinical Self-Assessment

Use Case ID	UC-04
Name	Complete a Clinical Self-Assessment (PHQ-9 / GAD-7)
Actors	End User (primary), Assessment Scoring Module (secondary)
Description	User takes a standardized questionnaire (PHQ-9 for depression or GAD-7 for anxiety), gets a scored result with severity interpretation, and the results are saved for tracking over time.
Trigger	User opens the Assessments section, or the system suggests it based on consistently low mood.
Preconditions	App is installed and onboarding done. User has not taken the same assessment in the last 24 hours.
Postconditions	Score is saved to user's profile and assessment history. If score indicates severe symptoms, crisis resources are shown.

Main Flow:

1. User taps Assessments from the menu.
2. System shows available assessments: PHQ-9 (Depression) and GAD-7 (Anxiety), with a short description and estimated time (2 to 3 minutes each).
3. User picks PHQ-9.
4. An intro screen explains this is a standard screening tool, not a diagnosis, and that answers stay on the device. User taps Begin.
5. System presents each question one at a time (9 questions total) with four options: Not at all (0), Several days (1), More than half the days (2), Nearly every day (3). A progress bar shows how far along the user is.
6. User answers all 9 questions.
7. System calculates the total score (0 to 27) and shows the result: the number, the severity level (Minimal 0-4, Mild 5-9, Moderate 10-14, Moderately Severe 15-19, Severe 20-27), what it means in plain language, and a comparison to their last score if they have one.
8. System shows a "What to do next" section with recommendations based on severity. For mild scores, it suggests continued tracking. For moderate and above, it encourages speaking to a professional.



9. User taps Done. Results are saved and they return to the assessments overview.

Alternative Flows:

A1: User picks GAD-7 instead. Same flow but with 7 anxiety-focused questions and scoring on a 0 to 21 scale.

A2: User checks assessment history. They tap History to see a timeline of past scores with a trend line showing improvement or decline.

A3: User quits midway. System asks if they are sure since progress will not be saved. If confirmed, partial answers are discarded.

Exception Flows:

E1: Severe score detected. If the score is 20+ or the suicide-related question (Q9) is answered with anything other than "Not at all," the system shows a resource card with crisis helpline numbers and a reminder that Moodify is not a substitute for professional help. The user must acknowledge this before seeing the regular results.

E2: Already taken today. System tells the user they completed this assessment recently and recommends waiting at least 2 weeks for accurate results. They can view their latest score but cannot retake it.



UC-05: Export Mood Data and Share with Therapist

Use Case ID	UC-05
Name	Export Mood Data and Share with Therapist
Actors	End User (primary), Therapist/Counselor (recipient), Export Module (secondary)
Description	User exports their mood data, activity logs, and assessment results as a PDF report or CSV file and shares it with their therapist via email or any other method.
Trigger	User taps Share with Therapist from Insights, or goes to Settings > Export Data.
Preconditions	At least 1 mood entry exists. Device has enough storage for generating the file.
Postconditions	PDF or CSV file generated and shared/saved. No data is modified or deleted by the export.

Main Flow:

1. User taps Share with Therapist or goes to Settings > Export Data.
2. Export config screen appears with options: date range (last 7 days, 30 days, 3 months, all time, or custom), data to include (mood entries, activities, emotions, notes, assessment results as checkboxes), and format (PDF Report or CSV).
3. User picks Last 30 days, checks everything, and selects PDF.
4. User taps Generate Report.
5. System generates a formatted PDF with a mood trend summary, a table of all entries, assessment scores if included, and a disclaimer that this is not a medical record.
6. System shows a preview of the report.
7. User taps Share to open the device's native share sheet and picks their preferred method (email, WhatsApp, etc.).
8. File is sent. System shows a confirmation and returns the user to the previous screen.



Alternative Flows:

A1: User picks CSV instead of PDF. System generates a CSV with columns for date, time, mood level, emotions, activities, notes, and assessment scores. Useful for therapists who want to run their own analysis.

A2: User saves to device instead of sharing. File goes to the Downloads folder and the system confirms with the file name.

A3: User wants to delete data after export. A Delete My Data option is available. System asks for confirmation since it is permanent and irreversible. If confirmed, all local data is wiped. This supports GDPR-style data rights.

A4: User exports only assessment history. From the Assessments section, they can tap Export Assessment History to get a focused report with just scores and trend charts, which is most useful for sharing with a psychiatrist.

Exception Flows:

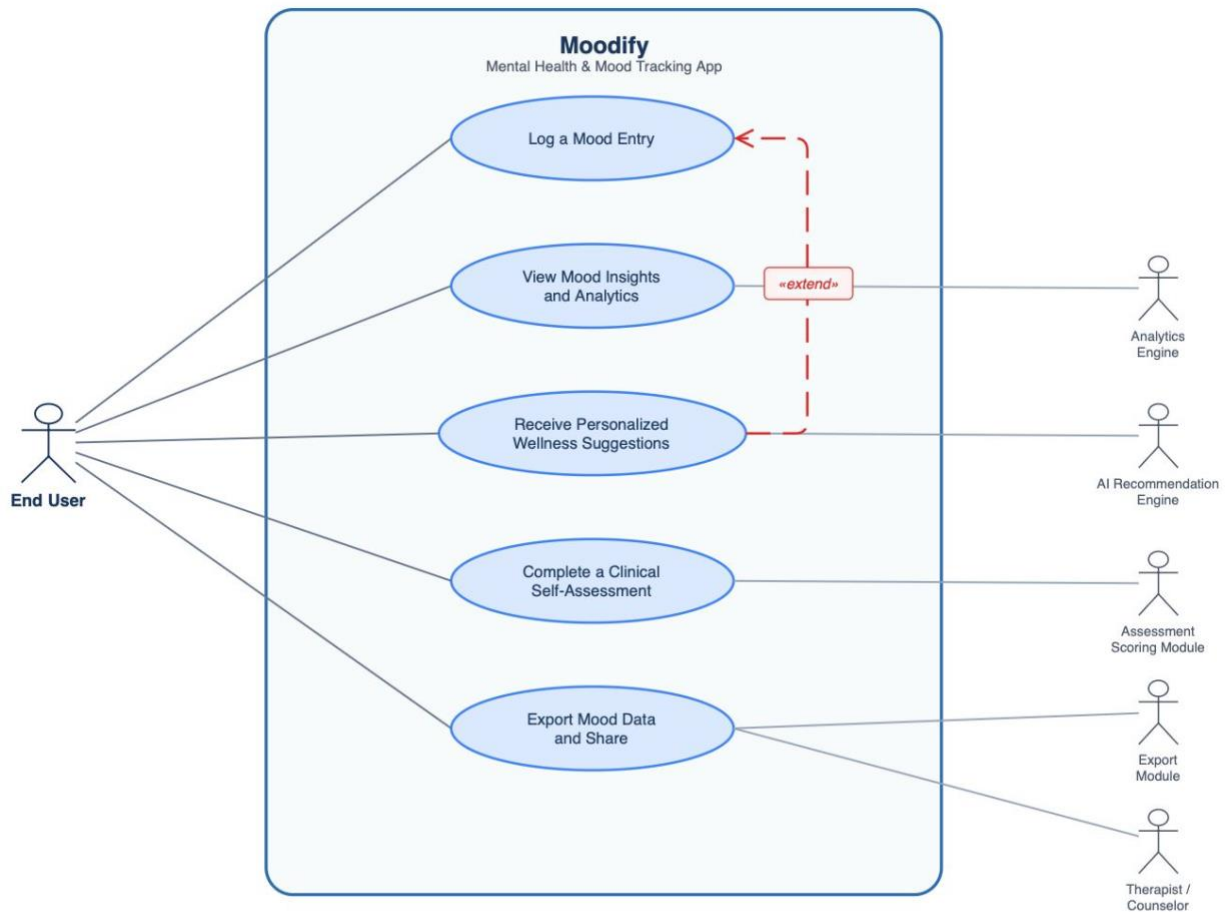
E1: No data in selected range. System tells the user and asks them to pick a different date range.

E2: PDF generation fails (e.g., not enough memory). System suggests trying a shorter date range or exporting as CSV instead.

E3: Share method fails (e.g., no email configured). The device handles the error natively, and the system offers to save locally as a fallback.

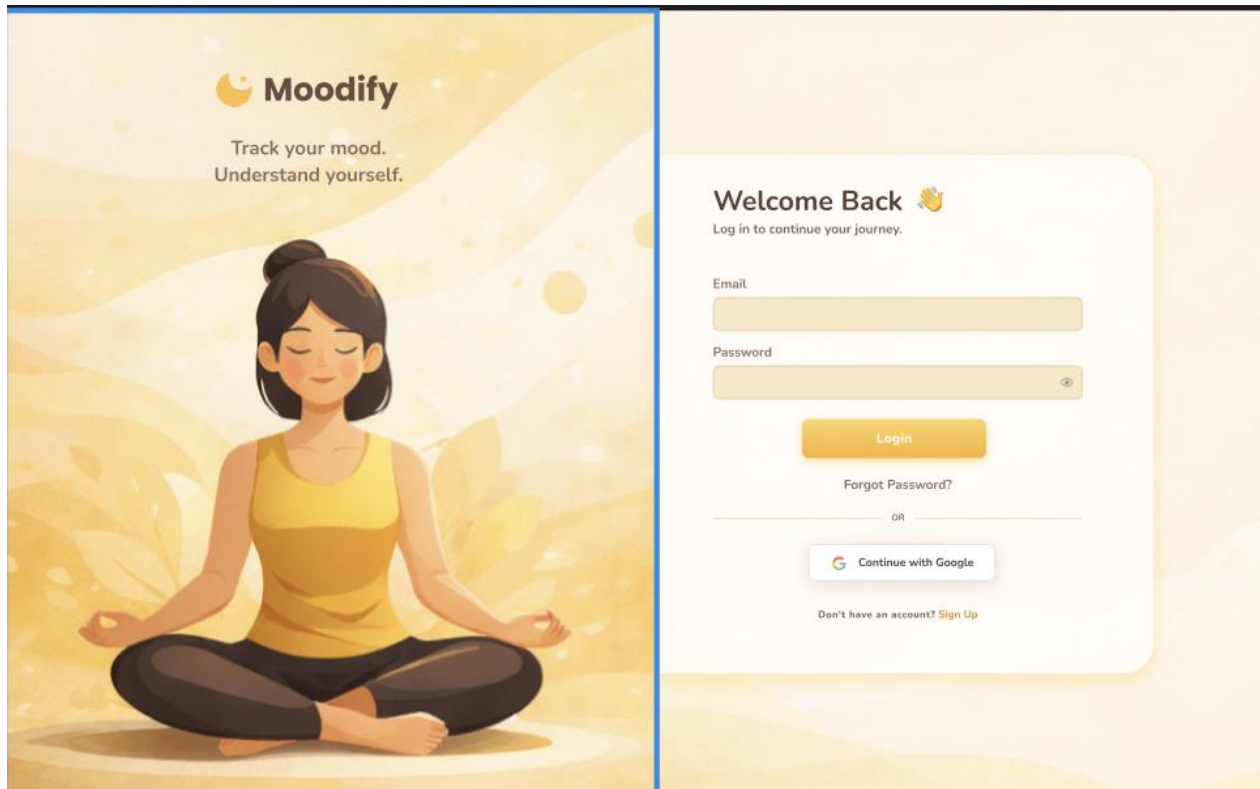


Use Case Diagram:



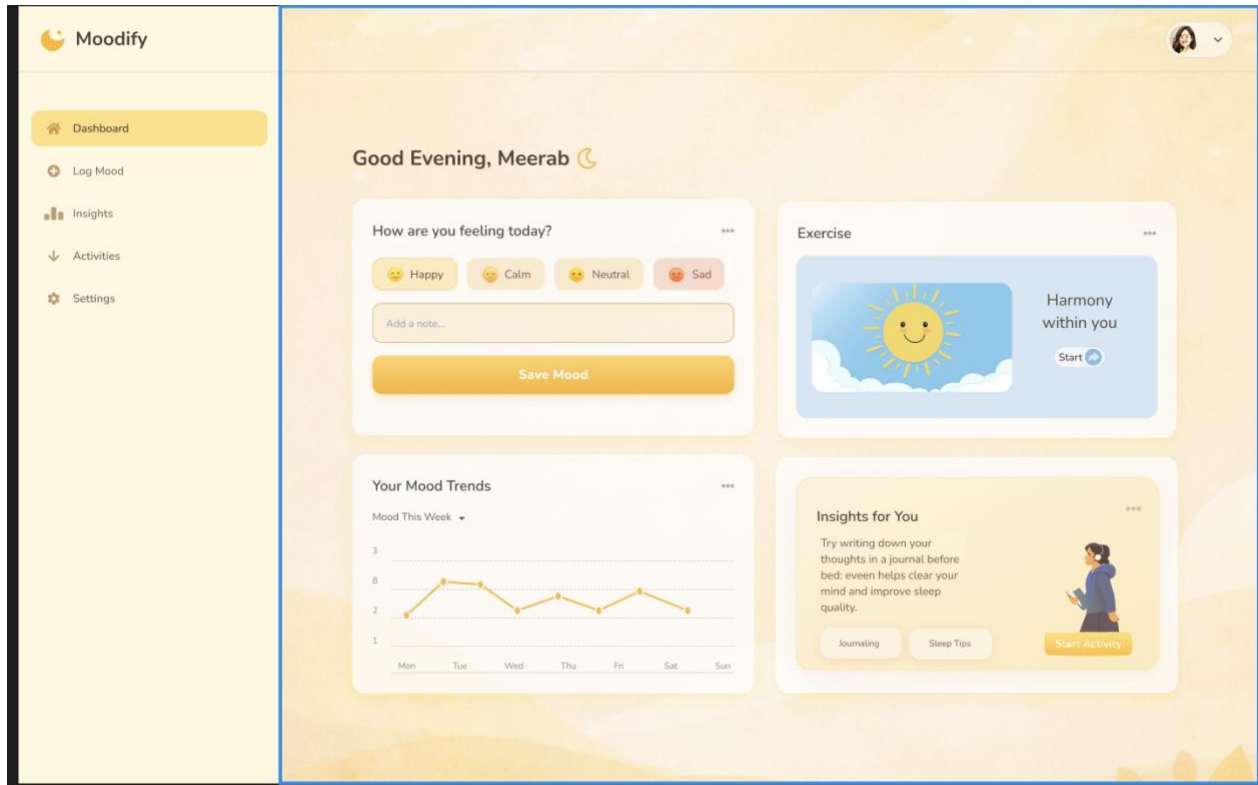


Figma Login Screen:





Figma Dashboard Screen:



Figma Link:

<https://www.figma.com/proto/6vdpLrYCWUPflvdUOinY3x/Untitled?node-id=9-56&p=f&t=v337jNvcRAasiwI4-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1>